

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #4

Instructions

- TA in charge: Sihui Shen (sihui.shen@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (10%)	Question 2 (25%)	Question 3 (10%)	Question 4 (05%)	Total
Question 1 (12%)	Question 2 (13%)	Question 3 (12%)	Question 4 (13%)	

Question 1 (10%)

Implement the function $h(x_1, x_2, x_3) = \sum m(1, 2, 3, 4, 5, 6)$ using a single 4-to-1 multiplexer. Assume that the complements of all input variables are given at no additional cost.

Question 2 (25%)

Design and synthesize the following system:

Inputs: $X = x_3x_2x_1x_0$, $Y = y_3y_2y_1y_0$, and $S = s_1s_0$

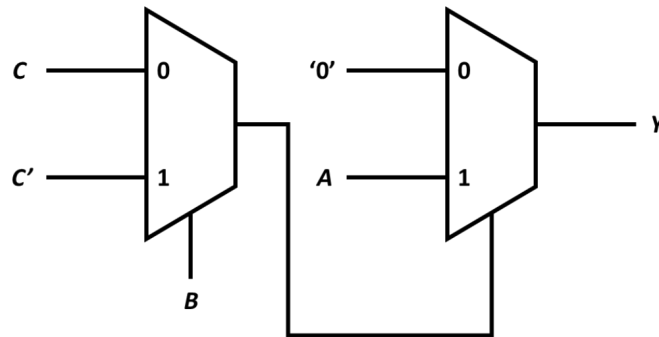
Output: Z

The functionality of the system is described in the table below. X and Y are signed numbers in 2's complement representation. Make sure to account for any possible overflow. You may use any gates and logic blocks learned in class, and the constants '0' and '1'.

s_1s_0	Functionality
00	$Z = X + Y$
01	$Z = X + 2 \cdot Y$
10	$Z = X - Y$
11	$Z = 0.5 \cdot X - Y$

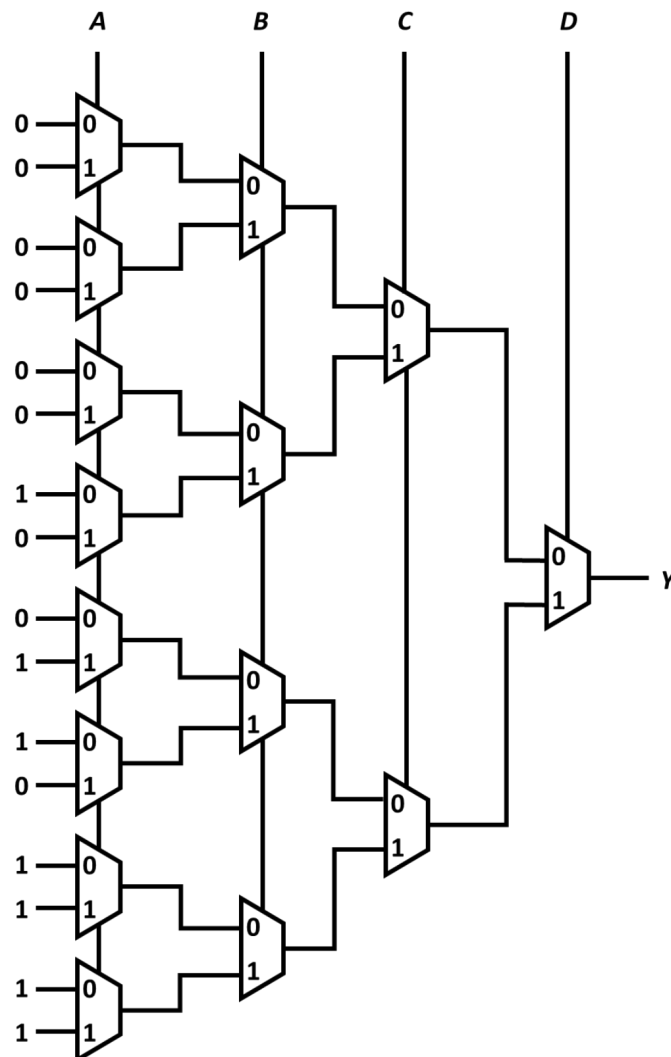
Question 3 (10%)

- (a) Implement a full adder using as few 4-1 MUXs as possible and the constants '0' and '1' (no other gates may be used). Assume that the complements of all input variables are given at no additional cost.
- (b) Determine the Boolean expression of the output Y in the following circuit:



Question 4 (05%)

Determine the canonical SOP expression of $Y(A, B, C, D)$, that is implemented by the following circuit.



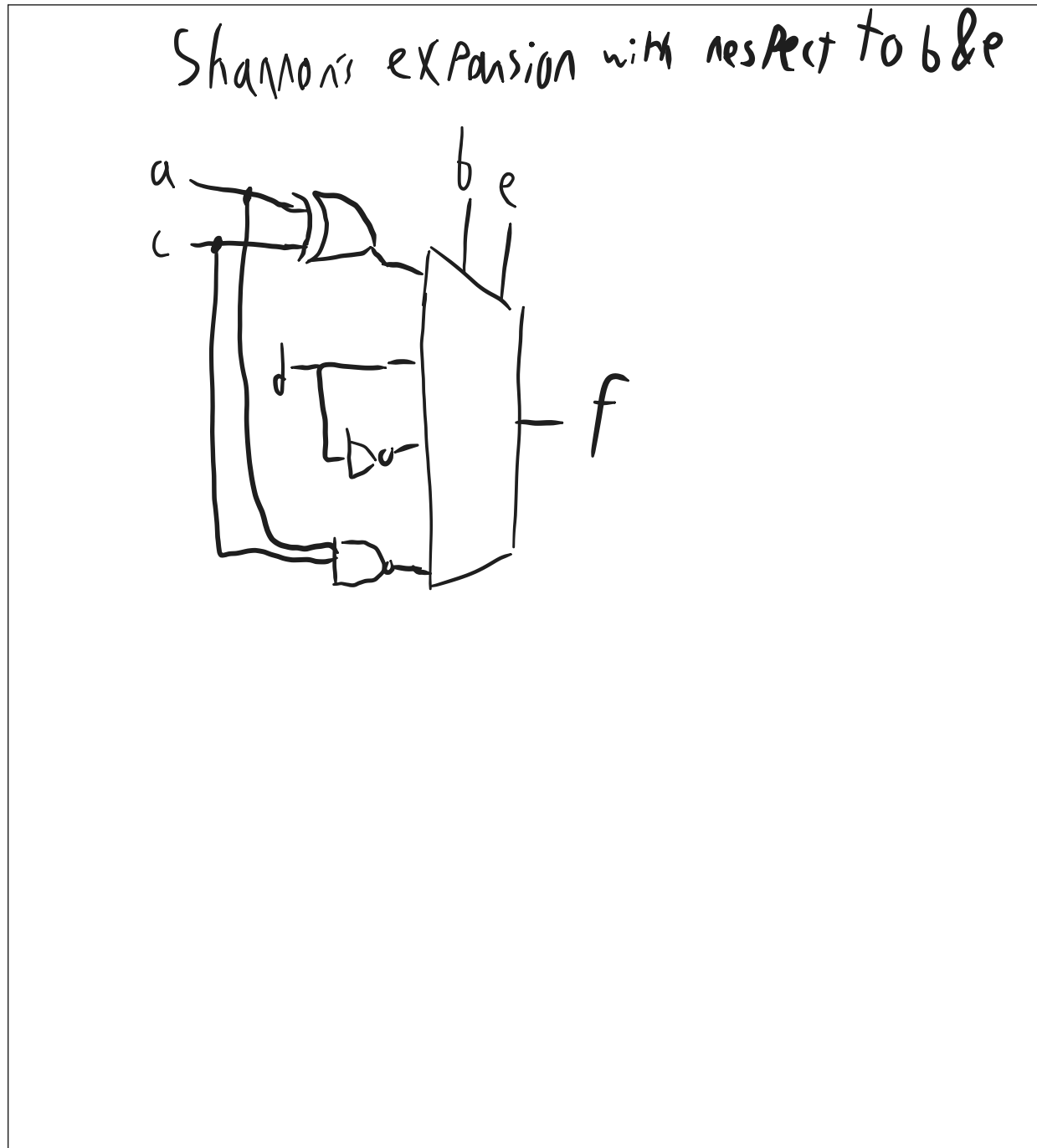
Question 5 (12%)

Implement the given function:

$$f(a, b, c, d, e) = abce + b\bar{d}\bar{e} + \bar{b}de + \bar{b}\bar{e}(a \oplus c) + \bar{a}b\bar{c}e$$

Use a single 4-to-1 multiplexer and as few other gates as possible. Assume that only the uncomplemented form of the inputs is available.

Hint: use Shannon's expansion with respect to variables that appear more often in the function.



Question 6 (13%)

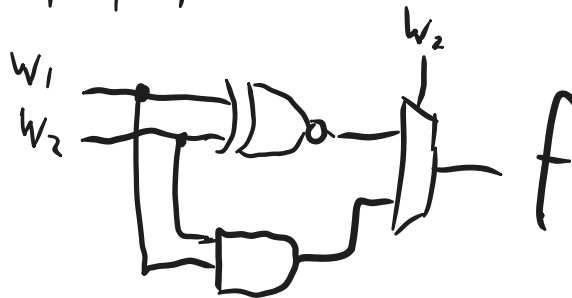
Use Shannon's expansion to implement the function $f(w_1, w_2, w_3) = \sum m(0, 5, 7)$ using 2-to-1 multiplexer and required gates.

	w_1	w_2	w_3	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

0 0 0
1 0 1
1 1 1

$$\overline{w_1} \overline{w_2} \overline{w_3} + w_1 \overline{w_2} w_3 + w_1 w_2 w_3$$

Shannon's expansion about w_2

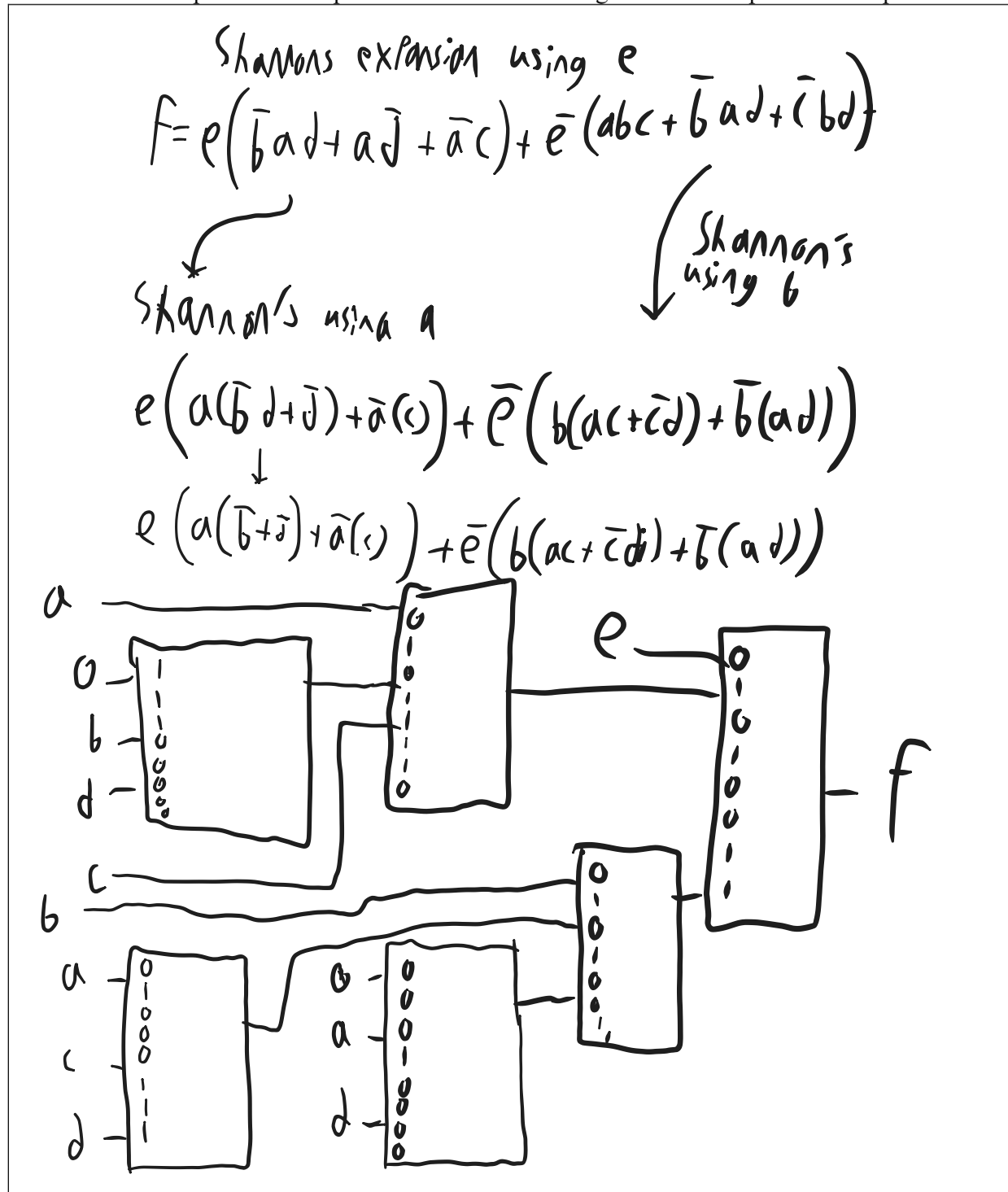


Question 7 (12%)

Given the following function:

$$f(a, b, c, d, e) = abc\bar{e} + \bar{b}ad + \bar{c}\bar{e}bd + a\bar{d}e + \bar{a}ce$$

Use Shannon's expansion to implement the function using as fewer 3-input LUTs as possible.



Question 8 (13%)

Design a comparator circuit that compares two 4-bit unsigned integers A and B. The circuit has three outputs: E equals '1' if $A = B$, G equals '1' if $A > B$ and is '0' otherwise, and L equals '1' if $A < B$ and is '0' otherwise. You can only use XOR and NOT gates, a priority encoder, and exactly two (as small as possible) multiplexers. Provide the circuit diagram of the comparator.

